Herbert Wertheim College of Engineering
UNIVERSITY of FLORIDA

Lecture 5F - Visual Search & Signal Detection Theory I

EIN3241: Human Factors and Ergonomics
Spring 2026

March 11, 2026

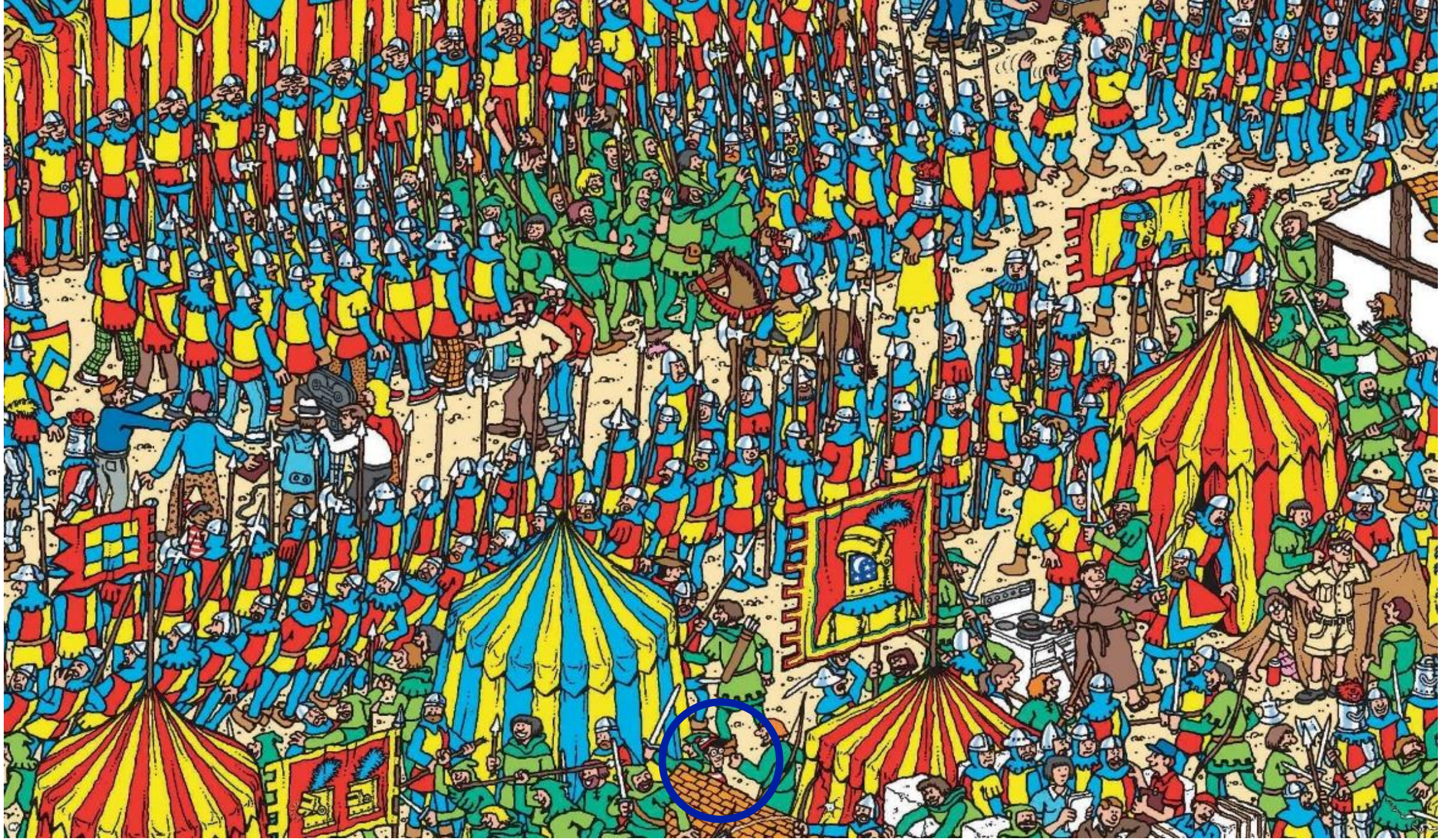
Dr. Jad Atweh

Admin Issues

- Lab 5 due on March 23

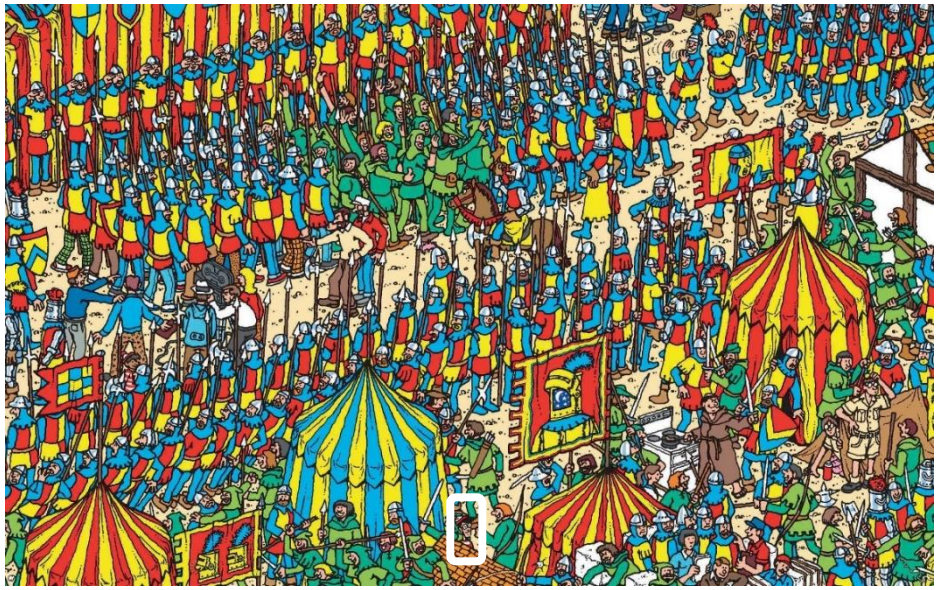
Learning objectives

- Understand human visual search and scanning
  - Serial search model
  - Conspicuity and expectancy
- Explain the core components of Signal Detection Theory and how they relate to decision-making under uncertainty.
- Apply Signal Detection Theory to analyze human performance in real-world tasks (e.g., medical diagnosis, security screening, or interface design).



Visual Search

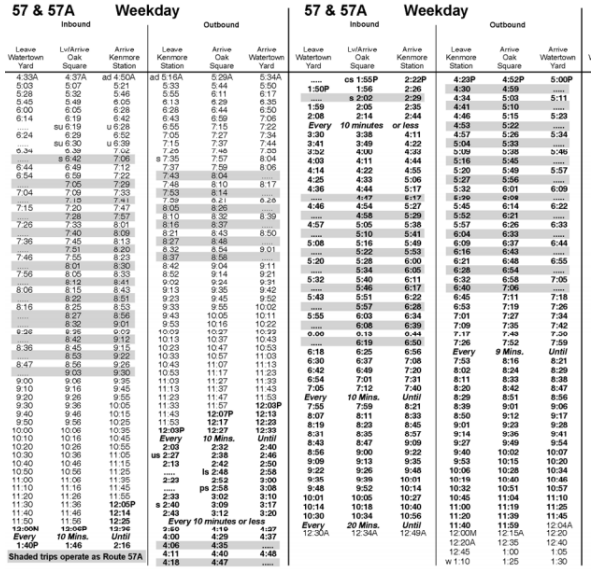
- Goal of visual search / scanning is to detect a target
  - e.g., “Where’s Waldo” (image search game)
- Object of interest = target = signal
- Things you do not care about = non-target = distractor = visual noise
- From an HFE perspective, noise impacts where and how objects should be presented.
  - Location of targets relative to “distracters”
  - Formatting of signal / stimulus for detection



Visual search examples

- Searching vehicle control panel for specific controls
  - Position is difficult to memorize due to:
    - Number of controls
    - Limited forms of tactile coding of controls

- Search for specific departure and arrival times on bus schedule
  - Similar distracters make target identification slow
  - Bold and highlighting can accelerate search



57 & 57A Weekday

Inbound: Leave Watertown Yard	Inbound: Lv/Arrive Oak Square	Inbound: Arrive Kenmore Station	Outbound: Leave Kenmore Station	Outbound: Arrive Oak Square	Outbound: Arrive Watertown Yard
4:33A	4:37A	ad 4:50A	ad 5:16A	5:29A	5:34A
5:03	5:07	5:21	5:33	5:44	5:50
5:28	5:32	5:46	5:55	6:11	6:17
5:48	5:48	6:05	6:13	6:29	6:35
6:00	6:05	6:28	6:28	6:44	6:50
6:14	6:19	6:42	6:43	6:59	7:06
....	su 6:19	u 6:28	6:55	7:15	7:22
6:24	6:29	6:52	7:05	7:27	7:34
	su 6:30	u 6:39	7:15	7:37	7:44
6:34	6:39	7:02	7:25	7:45	7:53
	s 6:42	7:06	s 7:35	7:57	8:04
6:44	6:49	7:12	7:37	7:59	8:06
6:54	6:59	7:22	7:43	8:04	
	7:05	7:29	7:48	8:10	8:17
7:04	7:09	7:33	7:53	8:14	
	7:15	7:41	7:59	8:21	8:28
7:15	7:20	7:47	8:05	8:26	
	7:28	7:57	8:10	8:32	8:39
7:26	7:33	8:01	8:16	8:37	
	7:40	8:09	8:21	8:43	8:50
7:36	7:45	8:13	8:27	8:48	
	7:51	8:20	8:32	8:54	9:01
7:46	7:55	8:23	8:37	8:58	
	8:01	8:30	8:42	9:04	9:11
7:56	8:05	8:33	8:52	9:14	9:21
	8:12	8:41	9:02	9:24	9:31
8:06	8:15	8:43	9:13	9:35	9:42
	8:22	8:51	9:23	9:45	9:52
8:16	8:25	8:53	9:33	9:55	10:02
	8:27	8:56	9:43	10:05	10:11
	8:32	9:01	9:53	10:16	10:22
8:26	8:35	9:03	10:03	10:27	10:33
	8:42	9:12	10:13	10:37	10:43
8:36	8:45	9:15	10:23	10:47	10:53
	8:53	9:22	10:33	10:57	11:03
8:47	8:56	9:26	10:43	11:07	11:13
	9:03	9:30	10:53	11:17	11:23
9:00	9:06	9:35	11:03	11:27	11:33
9:10	9:16	9:45	11:13	11:37	11:43
9:20	9:26	9:55	11:23	11:47	11:53
9:30	9:36	10:05	11:33	11:57	**12:03P**
9:40	9:46	10:15	11:43	**12:07P**	**12:13**
9:50	9:56	10:25	11:53	**12:17**	**12:23**
10:00	10:06	10:35	**12:03P**	**12:27**	**12:33**
10:10	10:16	10:45	***Every***	***10 Mins.***	***Until***
10:20	10:26	10:55	***2:03***	***2:32***	***2:40***
10:30	10:36	11:05	**us 2:27**	**2:38**	**2:46**
10:40	10:46	11:15	**2:13**	**2:42**	**2:50**
10:50	10:56	11:25	....	**ls 2:48**	**2:58**
11:00	11:06	11:35	**2:23**	**2:52**	**3:00**
11:10	11:16	11:45	....	**ps 2:58**	**3:08**
11:20	11:26	11:55	**2:33**	**3:02**	**3:10**
11:30	11:36	**12:06P**	**s 2:40**	**3:09**	**3:17**
11:40	11:46	**12:14**	**2:43**	**3:12**	**3:20**
11:50	11:56	**12:25**	***Every 10 minutes or less***		
12:00N	**12:06P**	**12:36**	**3:50**	**4:19**	**4:27**
Every	***10 Mins.***	***Until***	**4:00**	**4:29**	**4:37**
1:40P	**1:46**	**2:16**	**4:06**	**4:35**	....
			4:11	**4:40**	**4:48**
			4:18	**4:47**	....

Shaded trips operate as Route 57A

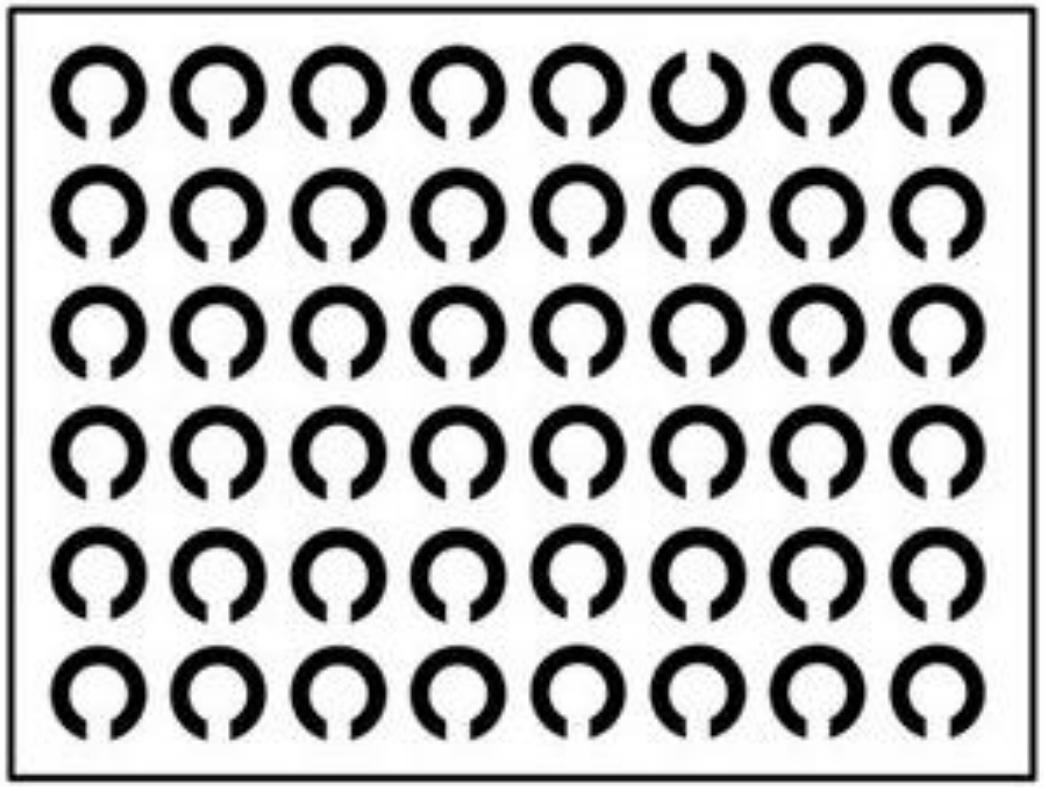
57 & 57A Weekday

Inbound: Leave Watertown Yard	Inbound: Lv/Arrive Oak Square	Inbound: Arrive Kenmore Station	Outbound: Leave Kenmore Station	Outbound: Arrive Oak Square	Outbound: Arrive Watertown Yard
....	**cs 1:55P**	**2:22P**	**4:23P**	**4:52P**	**5:00P**
1:50P	**1:56**	**2:26**	**4:30**	**4:59**	....
....	**s 2:02**	**2:29**	**4:34**	**5:03**	**5:11**
1:59	**2:05**	**2:35**	**4:41**	**5:10**	....
2:08	**2:14**	**2:44**	**4:46**	**5:15**	**5:23**
Every	***10 minutes***	***or less***	**4:53**	**5:22**	....
3:30	**3:38**	**4:11**	**4:57**	**5:26**	**5:34**
3:41	**3:49**	**4:22**	**5:04**	**5:33**	....
3:52	**4:00**	**4:33**	**5:09**	**5:38**	**5:46**
4:03	**4:11**	**4:44**	**5:16**	**5:45**	....
4:14	**4:22**	**4:55**	**5:20**	**5:49**	**5:57**
4:25	**4:33**	**5:06**	**5:27**	**5:56**	....
4:36	**4:44**	**5:17**	**5:32**	**6:01**	**6:09**
....	**4:47**	**5:17**	**5:39**	**6:08**	....
4:46	**4:54**	**5:27**	**5:45**	**6:14**	**6:22**
....	**4:58**	**5:29**	**5:52**	**6:21**	....
4:57	**5:05**	**5:36**	**5:57**	**6:26**	**6:33**
....	**5:10**	**5:41**	**6:04**	**6:33**	....
5:08	**5:16**	**5:49**	**6:09**	**6:37**	**6:44**
....	**5:22**	**5:53**	**6:16**	**6:43**	....
5:20	**5:28**	**6:00**	**6:21**	**6:48**	**6:55**
....	**5:34**	**6:06**	**6:28**	**6:54**	....
5:32	**5:40**	**6:11**	**6:32**	**6:58**	**7:05**
....	**5:46**	**6:17**	**6:40**	**7:06**	....
5:43	**5:51**	**6:22**	**6:45**	**7:11**	**7:18**
....	**5:57**	**6:28**	**6:53**	**7:19**	**7:26**
5:55	**6:03**	**6:34**	**7:01**	**7:27**	**7:34**
....	**6:08**	**6:39**	**7:09**	**7:35**	**7:42**
6:00	**6:13**	**6:44**	**7:17**	**7:43**	**7:50**
....	**6:19**	**6:50**	**7:26**	**7:52**	**7:59**
6:18	**6:25**	**6:56**	***Every***	***9 Mins.***	***Until***
6:30	**6:37**	**7:08**	**7:53**	**8:16**	**8:21**
6:42	**6:49**	**7:20**	**8:02**	**8:24**	**8:29**
6:54	**7:01**	**7:31**	**8:11**	**8:33**	**8:38**
7:05	**7:12**	**7:40**	**8:20**	**8:42**	**8:47**
Every	***10 Mins.***	***Until***	**8:29**	**8:51**	**8:56**
7:55	**7:59**	**8:21**	**8:39**	**9:01**	**9:06**
8:07	**8:11**	**8:33**	**8:50**	**9:12**	**9:17**
8:19	**8:23**	**8:45**	**9:01**	**9:23**	**9:28**
8:31	**8:35**	**8:57**	**9:14**	**9:36**	**9:41**
8:43	**8:47**	**9:09**	**9:27**	**9:49**	**9:54**
8:56	**9:00**	**9:22**	**9:40**	**10:02**	**10:07**
9:09	**9:13**	**9:35**	**9:53**	**10:15**	**10:20**
9:22	**9:26**	**9:48**	**10:06**	**10:28**	**10:34**
9:35	**9:39**	**10:01**	**10:19**	**10:40**	**10:46**
9:48	**9:52**	**10:14**	**10:32**	**10:51**	**10:57**
10:01	**10:05**	**10:27**	**10:45**	**11:04**	**11:10**
10:14	**10:18**	**10:40**	**11:00**	**11:19**	**11:25**
10:30	**10:34**	**10:56**	**11:20**	**11:39**	**11:45**
Every	***20 Mins.***	***Until***	**11:40**	**11:59**	12:04A
12:30A	12:34A	12:49A	12:00M	12:15A	12:20
			12:20A	12:35	12:40
			12:45	1:00	1:05
			w 1:10	1:25	1:30

Serial search model (Conjunction search)

- Search for target, inspect an object and make a decision, and then move on
- The idea is simple: You look at one object, decide if it's the target or not, and then move on to the next object
- When searching for target…
  - T = average time to find target
  - N = total number of items in search field
  - I = average inspection time for each item (assume constant)
  - On average you go through at least half items in visual field to find target (divide total field search time by 2)

$$T = \frac{N \times I}{2}$$

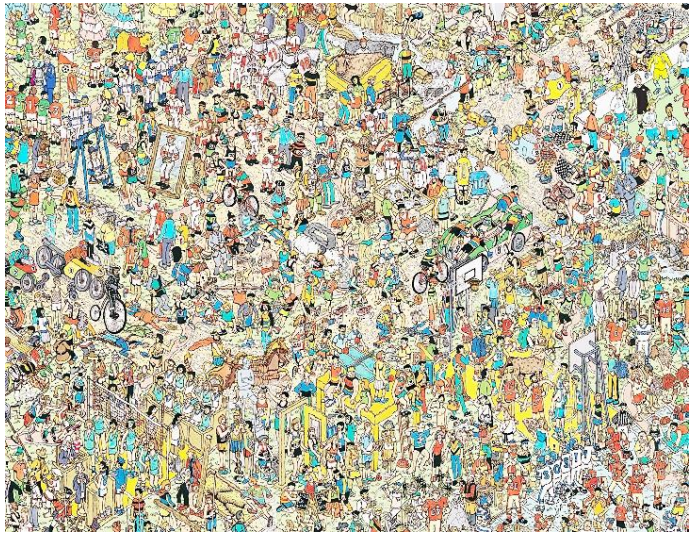


Matsuda, S., Matsumoto, H., Furubayashi, T., Fukuda, H., Emoto, M., Hanajima, R., ... & Terao, Y. (2014). Top-down but not bottom-up visual scanning is affected in hereditary pure cerebellar ataxia. *PloS one*, *9*(12), e116181.

Serial search model

- Systematic search in an organized environment (promotes efficiency):
  - From top to bottom
  - From left to right
  - Time has been found to follow linear model based on where you start and target location
- Unstructured search in a disorganized environment…
  - “Random” sampling of visual field
  - More time consuming

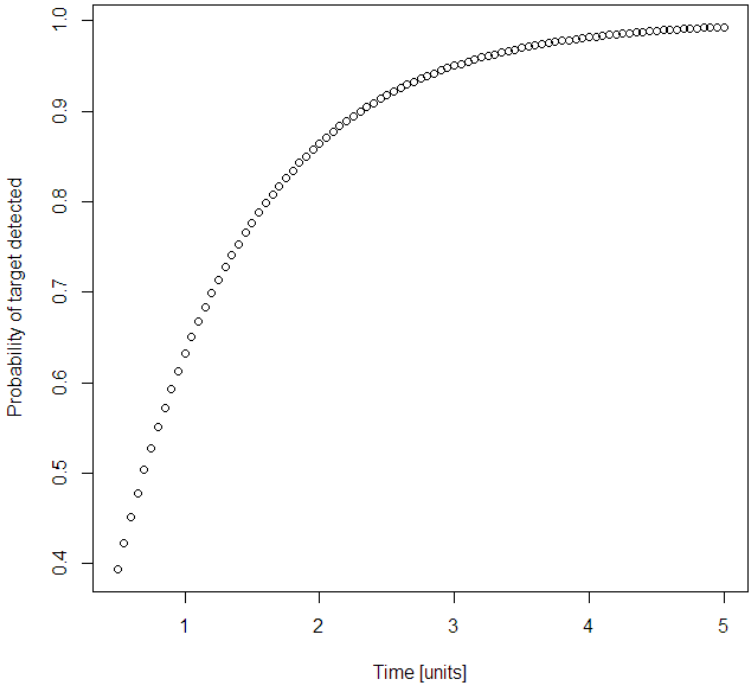
Law of “diminishing returns” applies in unstructured, random search process. Any increase in likelihood of target detection requires more and more task time.



School Word Search Puzzle

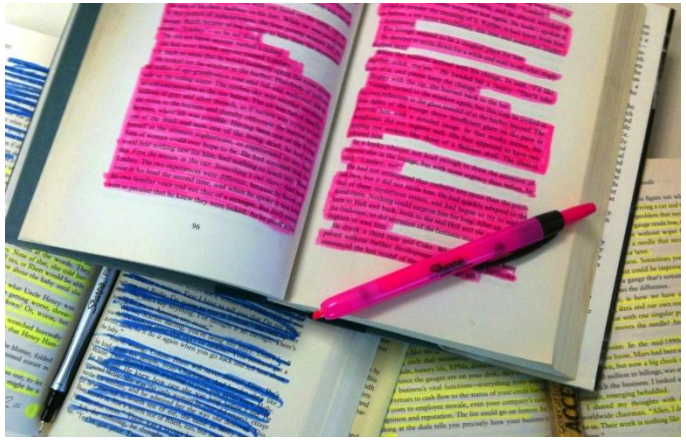
H	C	Y	A	S	S	E	Z	I	U	Q	H	P	Z	H
O	L	T	U	S	Z	G	A	M	E	S	O	F	B	V
M	Z	Z	N	S	E	S	S	A	L	C	M	W	U	H
E	D	E	T	E	N	T	I	O	N	Q	E	B	D	S
W	U	F	I	T	M	X	E	A	P	S	R	C	D	I
O	F	M	P	O	C	E	L	S	R	F	O	H	I	L
R	S	M	O	R	P	H	V	G	C	H	O	A	E	G
K	R	T	R	E	H	C	A	E	T	Z	M	L	S	N
B	E	M	U	L	E	A	R	N	I	N	G	K	E	E
E	A	J	T	D	L	L	I	R	D	H	T	B	C	G
L	D	H	S	P	E	L	L	I	N	G	C	O	N	T
L	I	T	E	N	G	N	S	K	D	W	Q	A	E	B
U	N	A	T	N	D	X	T	S	I	N	M	R	I	E
W	G	M	P	V	D	A	N	C	E	S	A	D	C	Y
L	A	P	I	C	N	I	R	P	A	B	E	B	S	J

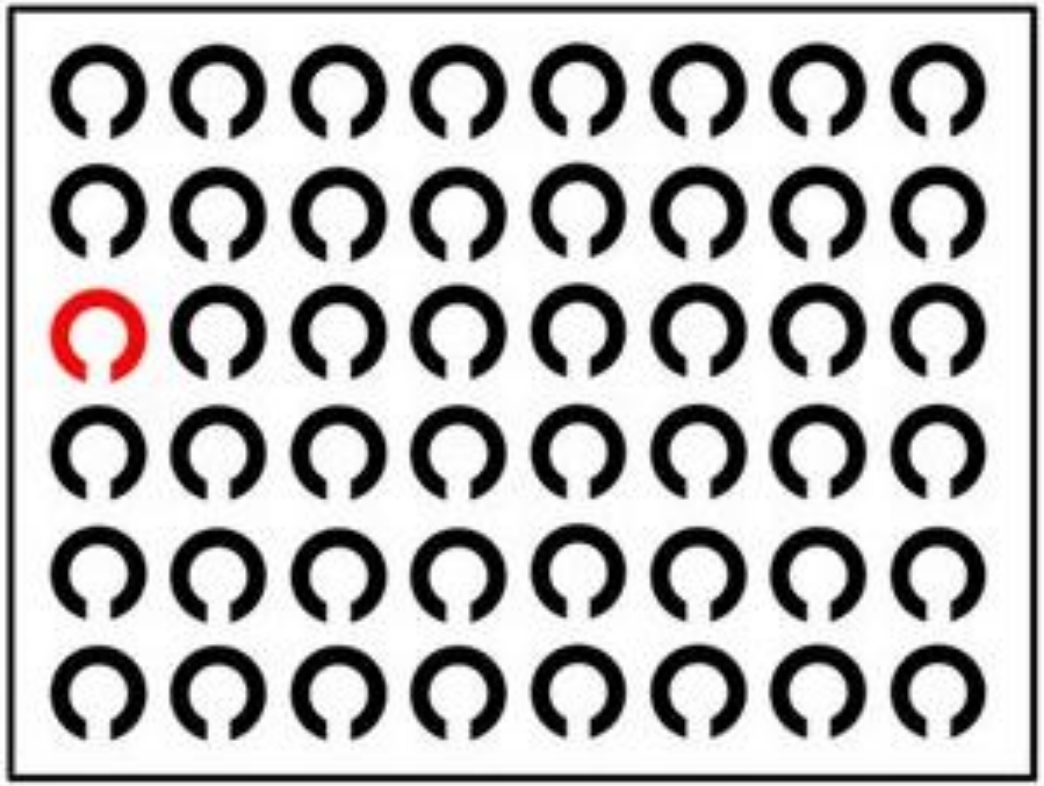
ACHIEVEMENT
BAND
BELL
BUDDIES
CHALKBOARD
CLASSES
DANCES
DETENTION
DRILL
ENGLISH
ESSAY
GAMES
HOMEROOM
HOMEWORK
LEARNING
MATH
PRINCIPAL
PROM
QUIZ
READING
SCIENCE
SPELLING
STUDENT
TEACHER
TEST



Conspicuity

- Conspicuity is how well targets stand out
- Promotes bottom-up processing (grabbing attention)
- Allows targets to be scanned in parallel (as opposed to in series; e.g., common coloring)
- Compromised when all targets are made to stand out!
- “Feature Search”, “Disjunctive Search”, “Efficient Search”





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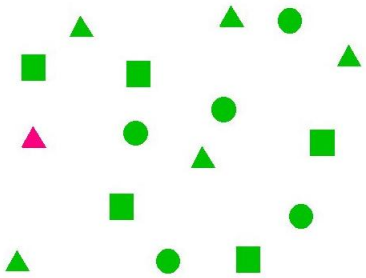
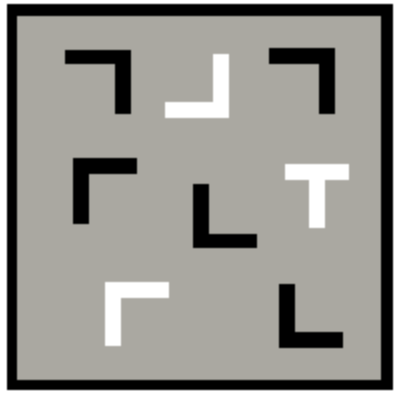
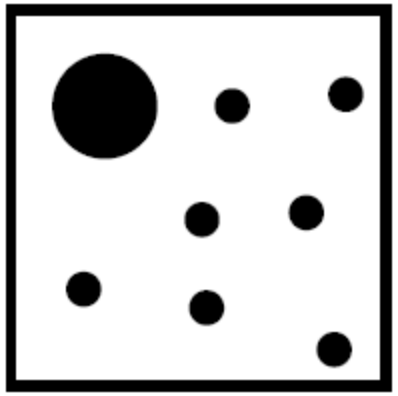
Properties affecting conspicuity

- Discriminability from background elements:
  - Coloring – Non-target (i.e., noise) items are uniformly colored
  - Size – Target is larger
  - Shape – Target is different
  - Brightness – Target is brighter
  - Motion – Background (i.e., noise) is stationary







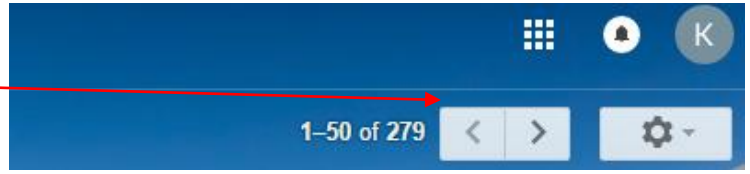
Properties affecting conspicuity

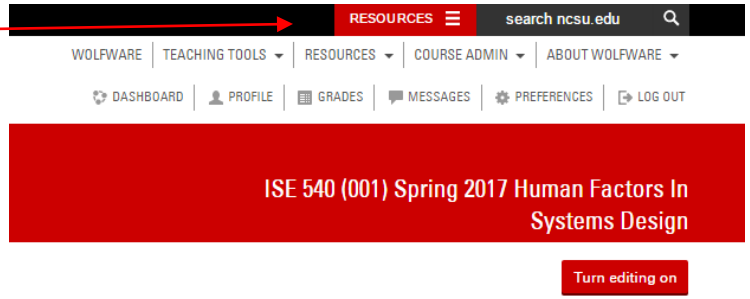
- Simplicity – Defining targets in terms of one feature (e.g., color=red) vs. multiple features (e.g., color=red and size=small)
- Automaticity - Target is highly familiar (e.g., your name) and stands out

Team logos with simple shapes and consistent colors draw saccadic response compared to very complex multi-feature coded logos

Human factors considerations

- Use knowledge of top-down processing to design interfaces that match user expectation / intuition
- Use knowledge of conspicuity to enhance visibility of targets in interface
- Use knowledge of serial search model (cost of information density and law of diminishing returns) to training users on strategy.
  - How to effectively search displays for targets (in absence of expectancy and conspicuity)





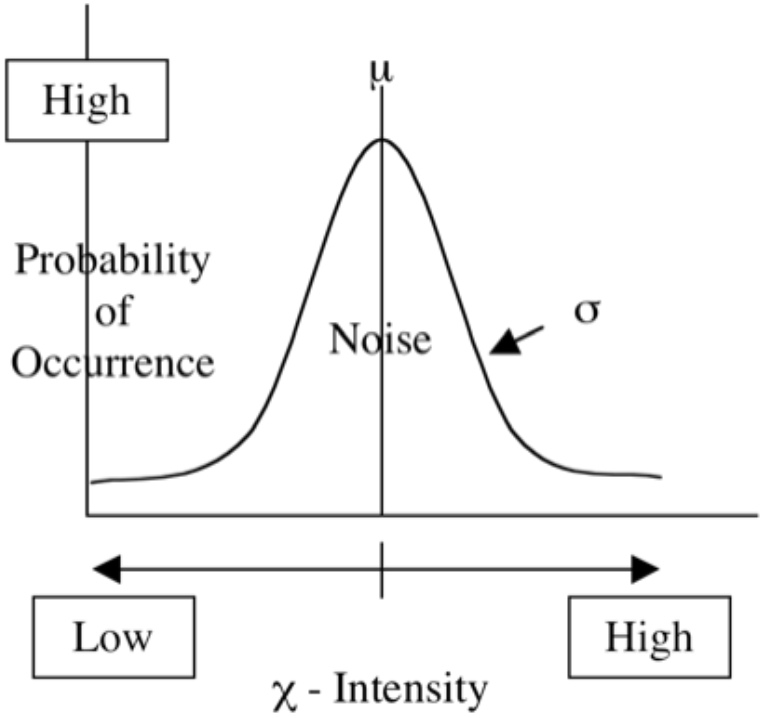
Signal detection theory

- Formal model of information and decision-making criteria
- Allows for optimal decision-making under certain conditions
- Provides a model of factors affecting human decision-making behavior

- History:
  - Original developed in late 1940s for radar detection problems
  - Extended to human perception and decision-making in 1960s
  - Used for complex human/automation alerting problems in the 1990s

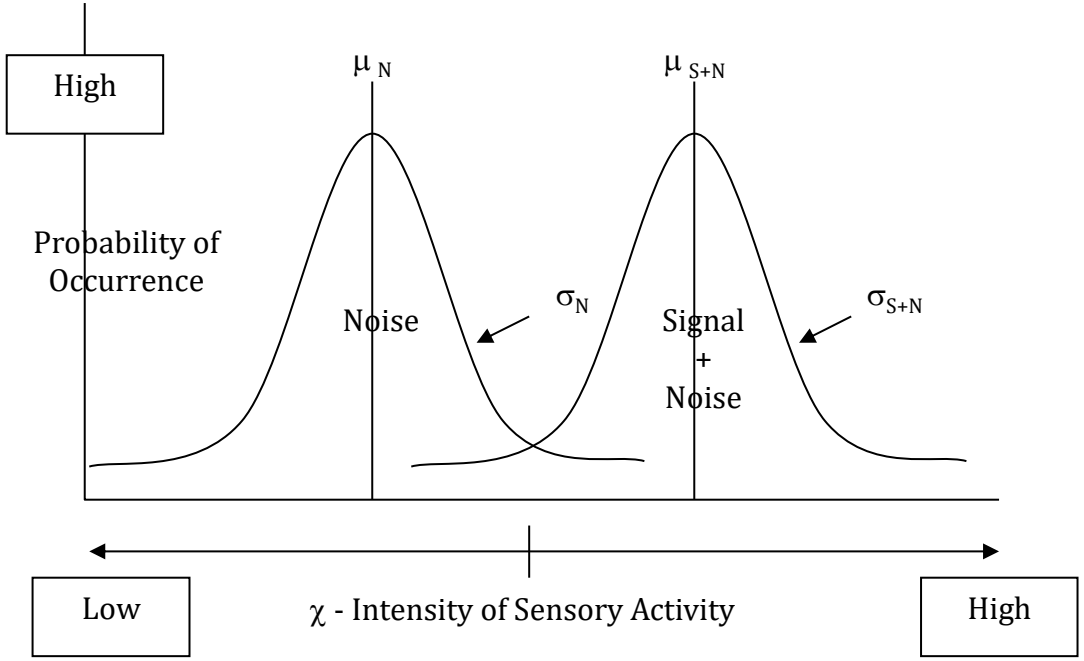
Signal detection theory

- Used to quantify ability to discern target/signal from noise:
  - e.g., medical lab technician scanning radiographs for tumor detection (present/absent)
- Concept of noise:
  - Varies with time and can be internal or external.
    - Internal – Misc. brain activity
    - External – Any stimulus other than target to be detected
  - Intensity assumed to follow normal distribution.



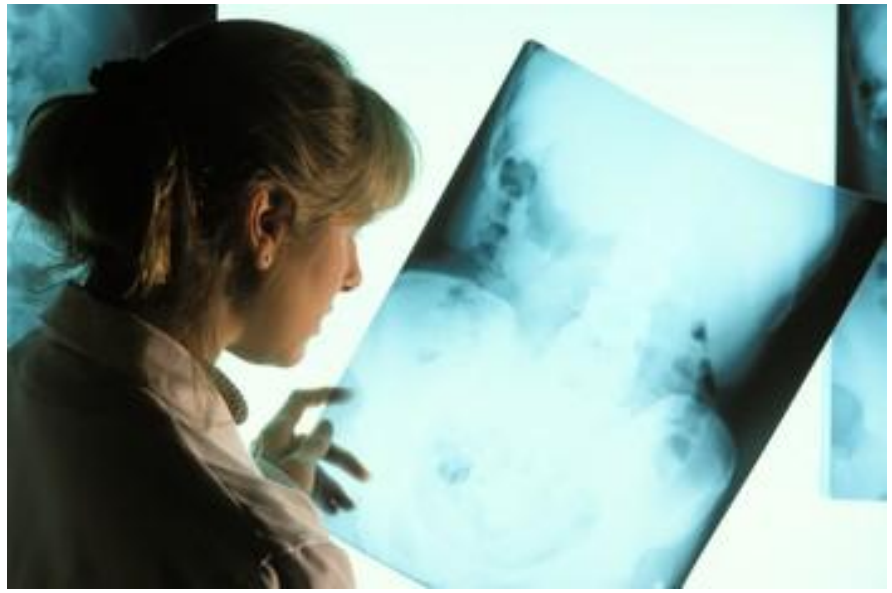
Signal added to noise

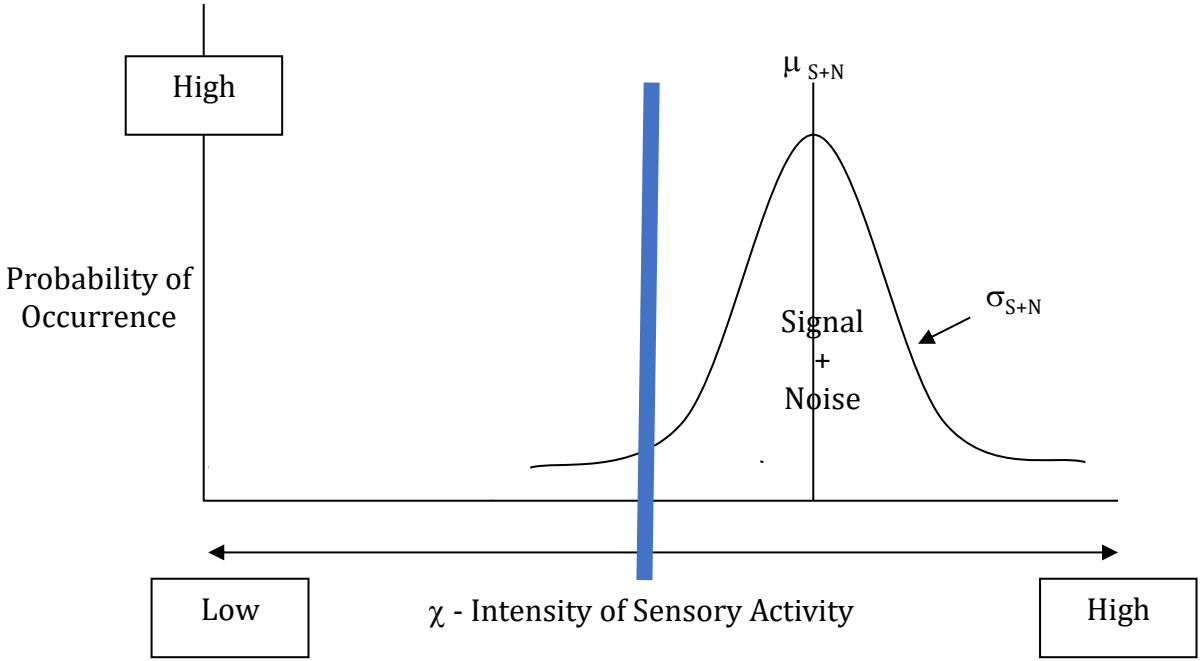
- Signal intensity may vary over time; also assumed to follow normal distribution.
- Possible that some noise may be more intense than signal + noise condition.
  - e.g., ATC operator using radar scope:
    - Moisture in air may mask signals from planes
    - Confusion of signal and noise is possible



Observer's Response

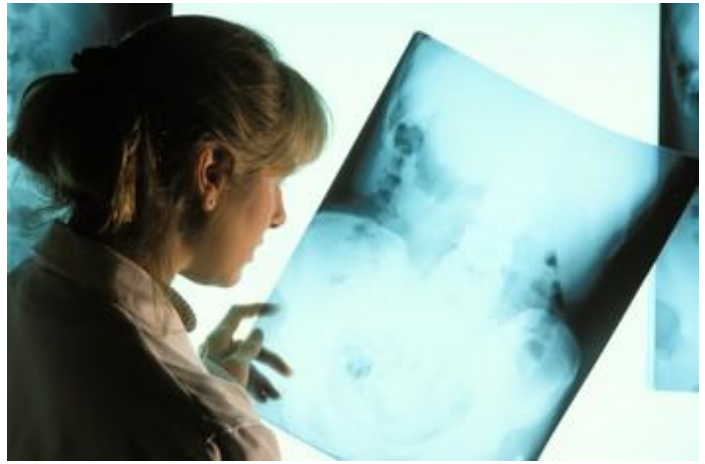
- Based on the stimuli the observer will have some level of "sensory activity" or "evidence"
- The observer doesn't know the true state of the world, but they need to make a choice, is this a signal?
  - Yes (it is signal + noise)
  - No (it is just noise)

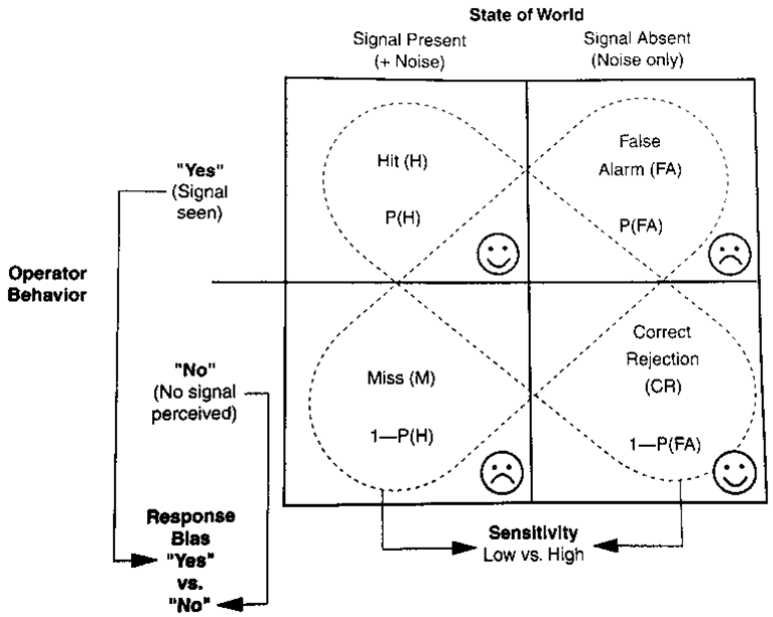




Signal detection theory

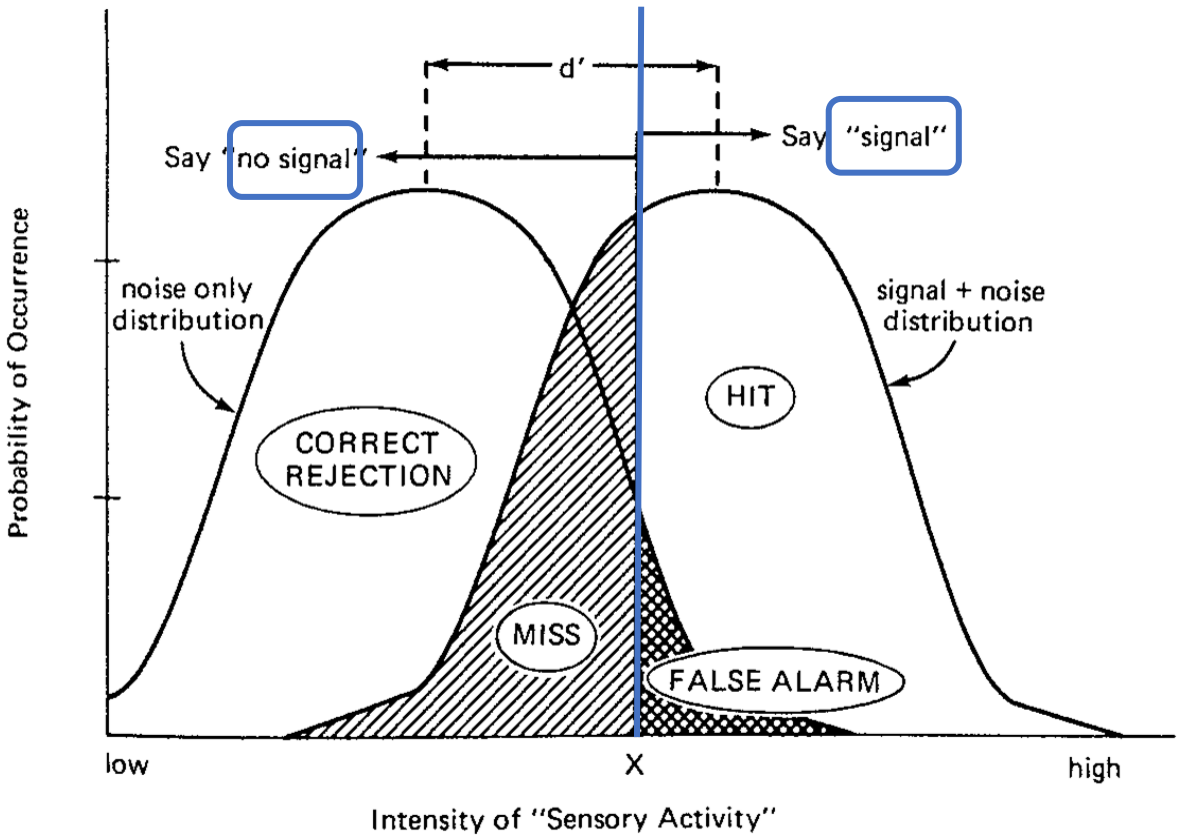
- States of reality – Noise or Signal + Noise
  - Categories of response – “Yes” or “No”
- Four possible outcomes in SD task:
  - Hit (H)
  - False alarm (FA)
  - Miss (M)
  - Correct rejection (CR)
- High **sensitivity** = high likelihood of H + CR
- Low **sensitivity** = high likelihood of FA + Miss





Signal detection theory

- Outcomes can be represented on overlapping PDFs of “noise” and “signal+noise”.
- Response criterion (X) defines likelihood of outcomes:
  - Level of stimulus intensity at which operator sets threshold for determining signal is present or not
- When reality=noise, only outcomes are FA & CR
- When reality=signal+noise, only outcomes are M + H

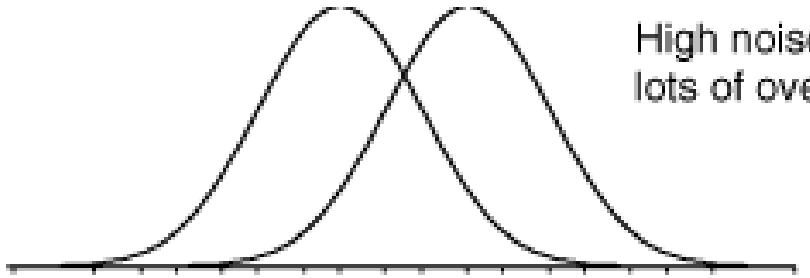


P(Noise) = P(FA) + P(CR) = 1

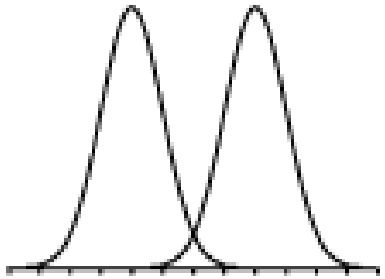
P(Signal+Noise) = P(H) + P(M) = 1

Sensitivity (discriminability)

- How hard / easy it is for observer to discriminate signal from noise
- Not influenced by response criterion; influenced by degree of separation between two distributions (noise and signal+noise)
  - Statistical distance between means of two distributions
- Greater separation = greater sensitivity
- Affects bottom-up processing – ability to detect signal based on features (e.g., brightness, loudness)
- Compromised by operator state - A fatigued operator is less sensitive; less capable of distinguishing signal from noise



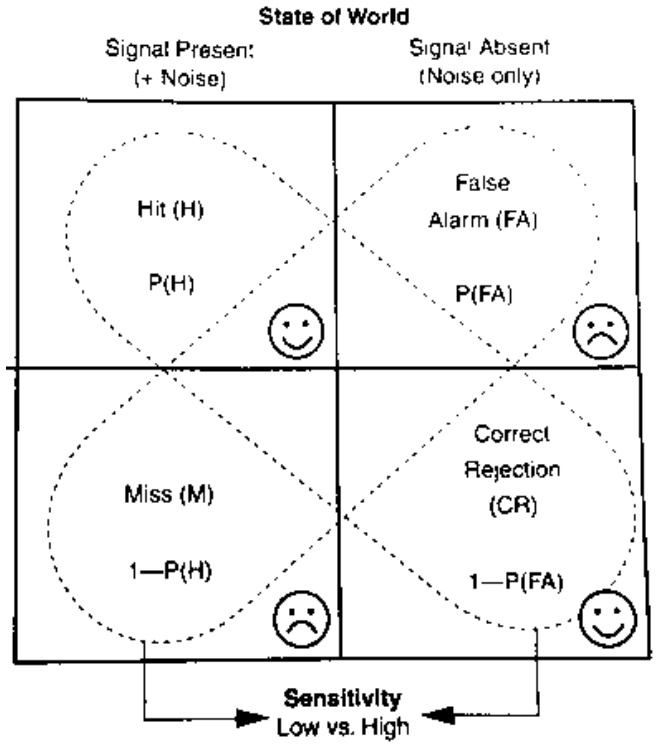
High noise,
lots of overlap

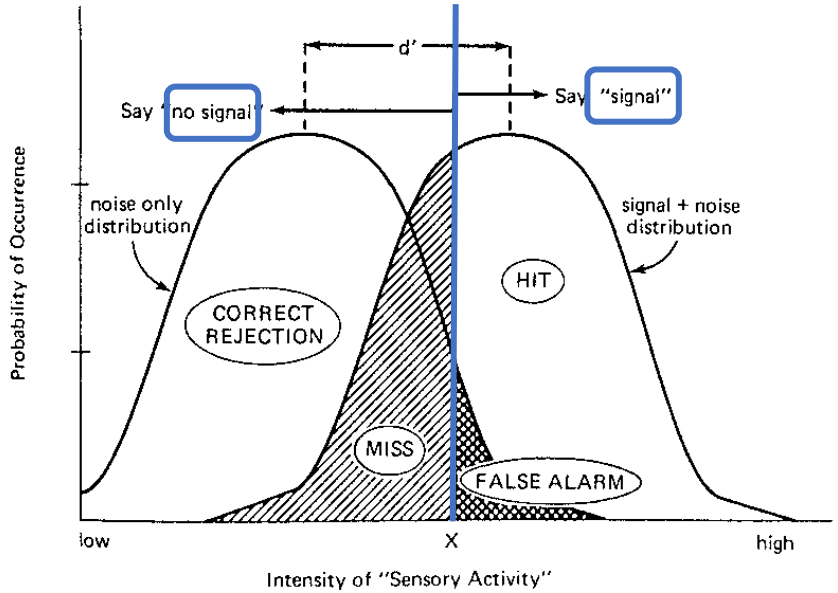


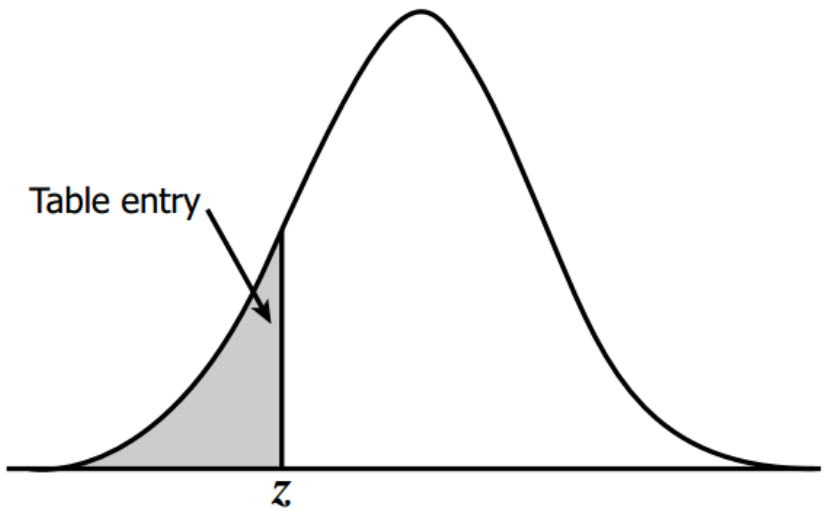
Low noise,
not much overlap

Sensitivity - discriminability (d')

- Can be measured as the proportion of correct responses:
  - $p(c) = \frac{\#\text{hits} + \#\text{correct rejections}}{\text{total \# of all possible outcomes}}$
- Can also be statistically determined:
  - z-score is based on standard normal dist.
  - $d' = Z(H) - Z(FA)$
- Calculation assumes equal variances for noise and signal+noise distributions



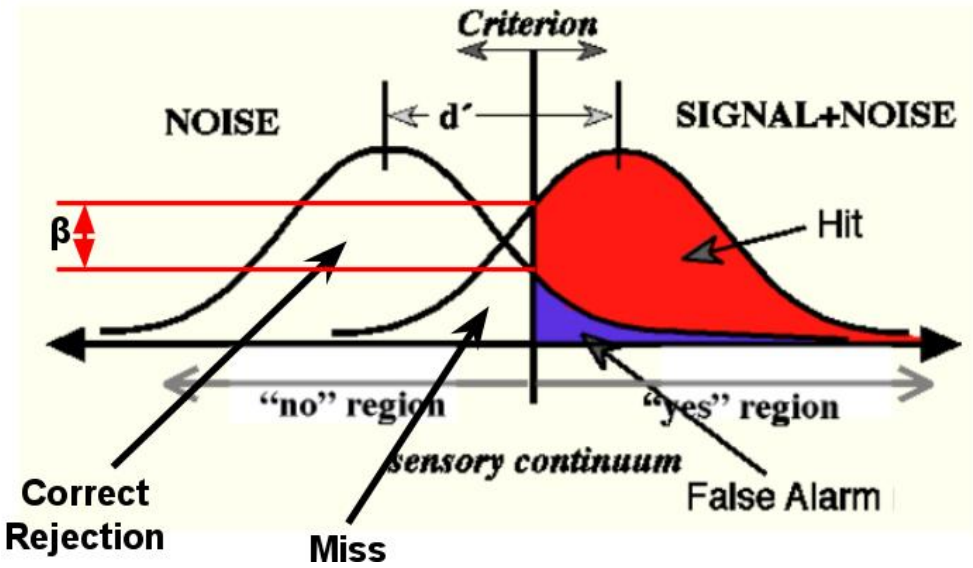




z	.00	.01	.02	.03	.04	.05	.06	.07	.08	.09
–3.4	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0003	.0002
–3.3	.0005	.0005	.0005	.0004	.0004	.0004	.0004	.0004	.0004	.0003
–3.2	.0007	.0007	.0006	.0006	.0006	.0006	.0006	.0005	.0005	.0005
–3.1	.0010	.0009	.0009	.0009	.0008	.0008	.0008	.0008	.0007	.0007
–3.0	.0013	.0013	.0013	.0012	.0012	.0011	.0011	.0011	.0010	.0010
–2.9	.0019	.0018	.0018	.0017	.0016	.0016	.0015	.0015	.0014	.0014
–2.8	.0026	.0025	.0024	.0023	.0023	.0022	.0021	.0021	.0020	.0019
–2.7	.0035	.0034	.0033	.0032	.0031	.0030	.0029	.0028	.0027	.0026
–2.6	.0047	.0045	.0044	.0043	.0041	.0040	.0039	.0038	.0037	.0036
–2.5	.0062	.0060	.0059	.0057	.0055	.0054	.0052	.0051	.0049	.0048
–2.4	.0082	.0080	.0078	.0075	.0073	.0071	.0069	.0068	.0066	.0064
–2.3	.0107	.0104	.0102	.0099	.0096	.0094	.0091	.0089	.0087	.0084
–2.2	.0139	.0136	.0132	.0129	.0125	.0122	.0119	.0116	.0113	.0110
–2.1	.0179	.0174	.0170	.0166	.0162	.0158	.0154	.0150	.0146	.0143
–2.0	.0228	.0222	.0217	.0212	.0207	.0202	.0197	.0192	.0188	.0183
–1.9	.0287	.0281	.0274	.0268	.0262	.0256	.0250	.0244	.0239	.0233
–1.8	.0359	.0351	.0344	.0336	.0329	.0322	.0314	.0307	.0301	.0294
–1.7	.0446	.0436	.0427	.0418	.0409	.0401	.0392	.0384	.0375	.0367
–1.6	.0548	.0537	.0526	.0516	.0505	.0495	.0485	.0475	.0465	.0455
–1.5	.0668	.0655	.0643	.0630	.0618	.0606	.0594	.0582	.0571	.0559
–1.4	.0808	.0793	.0778	.0764	.0749	.0735	.0721	.0708	.0694	.0681
–1.3	.0968	.0951	.0934	.0918	.0901	.0885	.0869	.0853	.0838	.0823
–1.2	.1151	.1131	.1112	.1093	.1075	.1056	.1038	.1020	.1003	.0985
–1.1	.1357	.1335	.1314	.1292	.1271	.1251	.1230	.1210	.1190	.1170
–1.0	.1587	.1562	.1539	.1515	.1492	.1469	.1446	.1423	.1401	.1379
–0.9	.1841	.1814	.1788	.1762	.1736	.1711	.1685	.1660	.1635	.1611
–0.8	.2119	.2090	.2061	.2033	.2005	.1977	.1949	.1922	.1894	.1867
–0.7	.2420	.2389	.2358	.2327	.2296	.2266	.2236	.2206	.2177	.2148
–0.6	.2743	.2709	.2676	.2643	.2611	.2578	.2546	.2514	.2483	.2451
–0.5	.3085	.3050	.3015	.2981	.2946	.2912	.2877	.2843	.2810	.2776
–0.4	.3446	.3409	.3372	.3336	.3300	.3264	.3228	.3192	.3156	.3121
–0.3	.3821	.3783	.3745	.3707	.3669	.3632	.3594	.3557	.3520	.3483
–0.2	.4207	.4168	.4129	.4090	.4052	.4013	.3974	.3936	.3897	.3859
–0.1	.4602	.4562	.4522	.4483	.4443	.4404	.4364	.4325	.4286	.4247
–0.0	.5000	.4960	.4920	.4880	.4840	.4801	.4761	.4721	.4681	.4641

Response Criterion - Beta (β)

- Quantifies the position of response criterion (X) along continuum of stimulus intensity
  - Beta = β = P (X|S+N) / P (X|N) (at criterion point)
- Ratio of ordinates of signal+noise curve over noise curve at response criterion point
  - Beta ≈ $\frac{\text{Height of signal+noise PDF at X}}{\text{Height of noise PDF at X}}$



= Hit $_{\text{ordinate}}$ / False Alarm $_{\text{ordinate}}$

= Ordinate(Z(p(H))) / Ordinate(Z(p(FA)))

$= e^{(0.5\,[Z(p(FA))^2 - Z(p(H))^2])}$

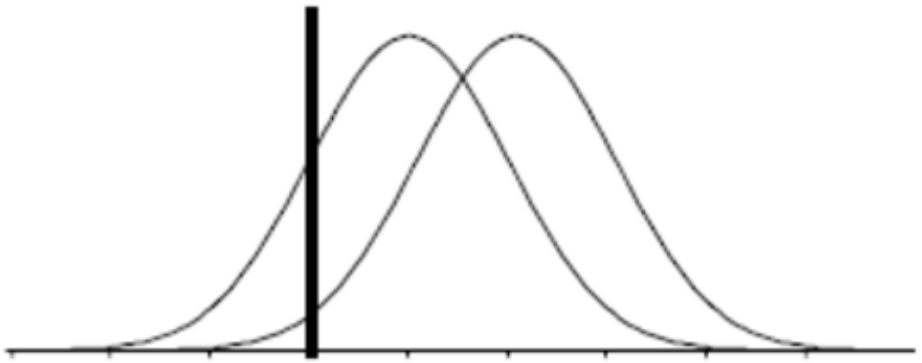
Normal Ordinates

$$\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}, \text{ for } 0 \le z \le 4.99, \phi(-z) = \phi(z)$$

z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.3989	0.3989	0.3989	0.3988	0.3986	0.3984	0.3982	0.3980	0.3977	0.3973
0.1	0.3970	0.3965	0.3961	0.3956	0.3951	0.3945	0.3939	0.3932	0.3925	0.3918
0.2	0.3910	0.3902	0.3894	0.3885	0.3876	0.3867	0.3857	0.3847	0.3836	0.3825
0.3	0.3814	0.3802	0.3790	0.3778	0.3765	0.3752	0.3739	0.3725	0.3712	0.3697
0.4	0.3683	0.3668	0.3653	0.3637	0.3621	0.3605	0.3589	0.3572	0.3555	0.3538
0.5	0.3521	0.3503	0.3485	0.3467	0.3448	0.3429	0.3410	0.3391	0.3372	0.3352
0.6	0.3332	0.3312	0.3292	0.3271	0.3251	0.3230	0.3209	0.3187	0.3166	0.3144
0.7	0.3123	0.3101	0.3079	0.3056	0.3034	0.3011	0.2989	0.2966	0.2943	0.2920
0.8	0.2897	0.2874	0.2850	0.2827	0.2803	0.2780	0.2756	0.2732	0.2709	0.2685
0.9	0.2661	0.2637	0.2613	0.2589	0.2565	0.2541	0.2516	0.2492	0.2468	0.2444
1.0	0.2420	0.2396	0.2371	0.2347	0.2323	0.2299	0.2275	0.2251	0.2227	0.2203
1.1	0.2179	0.2155	0.2131	0.2107	0.2083	0.2059	0.2036	0.2012	0.1989	0.1965
1.2	0.1942	0.1919	0.1895	0.1872	0.1849	0.1826	0.1804	0.1781	0.1758	0.1736
1.3	0.1714	0.1691	0.1669	0.1647	0.1626	0.1604	0.1582	0.1561	0.1539	0.1518
1.4	0.1497	0.1476	0.1456	0.1435	0.1415	0.1394	0.1374	0.1354	0.1334	0.1315
1.5	0.1295	0.1276	0.1257	0.1238	0.1219	0.1200	0.1182	0.1163	0.1145	0.1127
1.6	0.1109	0.1092	0.1074	0.1057	0.1040	0.1023	0.1006	0.09893	0.09728	0.09566
1.7	0.09405	0.09246	0.09089	0.08933	0.08780	0.08628	0.08478	0.08329	0.08183	0.08038
1.8	0.07895	0.07754	0.07614	0.07477	0.07341	0.07206	0.07074	0.06943	0.06814	0.06687
1.9	0.06562	0.06438	0.06316	0.06195	0.06077	0.05959	0.05844	0.05730	0.05618	0.05508
2.0	0.05399	0.05292	0.05186	0.05082	0.04980	0.04879	0.04780	0.04682	0.04586	0.04491
2.1	0.04398	0.04307	0.04217	0.04128	0.04041	0.03955	0.03871			

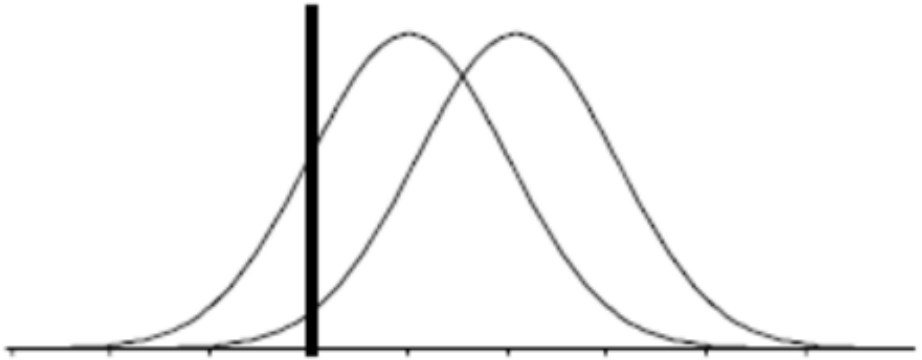
Response criterion shift

- Influencing response criterion (in simple detection task)
- If an operator knows that the likelihood of signals occurring is high…
  - As *p*(signal) (the probability of a signal occurring) increases, what happens to X (response criterion)?
    - Shifts to the **left**/right on continuum of activity
    - Operator is more likely to say **yes**/no to signal
    - Operator is more likely to say **yes**/no to false alarm



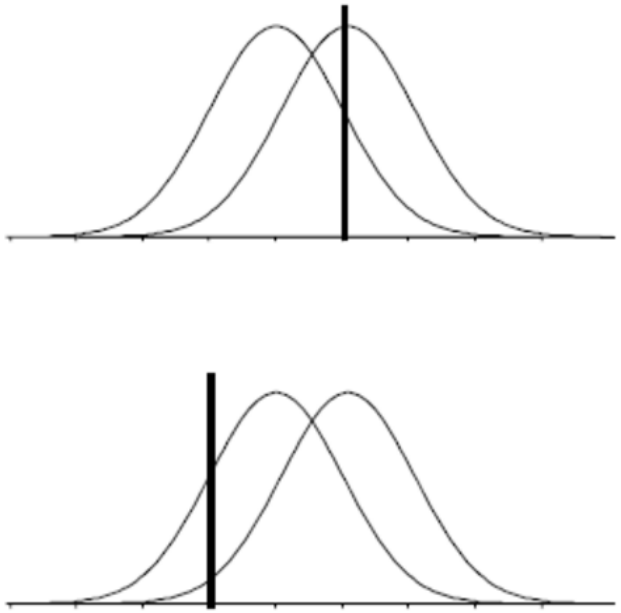
Response criterion shift

- **Example**: If field intelligence indicated that a terrorist had threatened a flight, thus increasing the likelihood of a threat object, the screener would be more likely to regard an alarmed bag as a threat object. That is, as the **probability of a signal increases**, the response criterion is lowered (**reducing β**), so that anything remotely suggesting a signal (a threat object in our example) might be **called a signal** (Sanders & McCormick, 1987).



Response criterion shift

- Influencing response criterion (in simple detection task)
  - **Cost of consequence of miss versus false alarm**:
    - If cost of false alarm is very high, what happens to position of decision criterion on continuum of sensory activity? Shifts to _____ to the right _____
      - Operator becomes more _____ conservative _____
      - Operator is more likely to say signal is _____ "No" _____
    - If cost of miss is high, what happens to position of decision criterion? Shifts to _____ to the left _____
      - Responder becomes _____ risky _____
      - Responder is more likely to say signal is _____ "Yes" _____



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LEADING THE CHARGE, CHARGING AHEAD